

TOI-1811b

R-2307

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI1811_240222 · created 2026-08-17T23:33:06+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460362.8224 +/- 0.0055 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.016
Transit depth (Rp/Rs)^2	2.15 +/- 0.46 [%]
Orbital Inclination (inc)	86.24 +/- 1.06
Airmass coefficient 1 (a1)	1.214 +/- 0.013
Airmass coefficient 2 (a2)	-0.134 +/- 0.024
Scatter in the residuals of the lightcurve fit is	1.06 %
Best Comparison Star	None
Optimal Aperture	3.8
Optimal Annulus	7.89
Transit Duration (day)	0.071 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.147 ± 0.016 vs 0.13272 (deviation 0.87σ)
Duration fit vs published	0.071 d vs 0.0813 d (deviation 0.4σ)
Mid-time O–C	4.0 min
Depth SNR	9.0
Residual scatter	1.06 %

Light curve

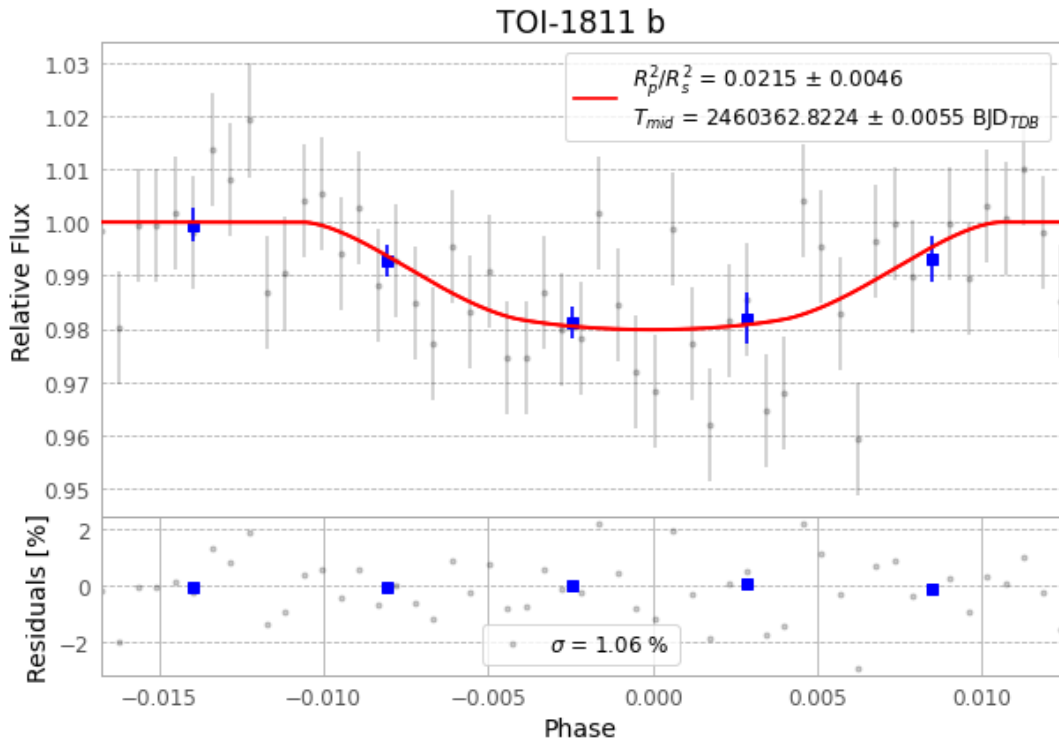


Figure 1 — FinalLightCurve_TOI-1811 b_2024-02-21.png (SHA-256 4f6fa0e13951b9c8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [396, 238], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0564f0e22a100fee...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

53 FITS frames (35 MB), TOI1811240222060616.FITS → TOI1811240222084216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1811-240222.log (cbf701a462f9...), FinalLightCurve_TOI-1811 b_2024-02-21.png (4f6fa0e13951...), FinalLightCurve_TOI-1811 b_2024-02-21.csv (99c2026be0e6...)

Compute resources

Wall time 112.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 23:33Z.